

Group, Ring & Field

Group: A group G sometimes denoted by $\{G, \cdot\}$ is a set of elements with binary operation denoted by $\cdot$ that associates to each ordered pair (a, b) of elements in G an element $(a \cdot b)$ in G , such that the following properties are obeyed.

Properties

Closure: If a, b belong to G , then $a \cdot b$ is also in G

Commutative: $a \cdot b = b \cdot a \quad \forall a, b \text{ in } G$

Associative: $a \cdot (b \cdot c) = (a \cdot b) \cdot c \quad \forall a, b, c \text{ in } G$

Identity: $a \cdot e = e \cdot a = a \quad \forall a \text{ in } G$

Inverse: $a \cdot a^{-1} = a^{-1} \cdot a = e$

Ring: It is an algebraic structure consisting of a set R with two operations addition $(+)$ and multiplication $(*)$.
A Ring sometimes denoted by $\{R, +, \times\}$.

Properties

1. Closure: for all $a, b \in R$, both $a+b$ and $a \times b$ are in R

2. Addition forms an abelian group:

Associative: $a + (b + c) = (a + b) + c$

Commutative: $a + b = b + a$

Identity: $a + 0 = a$

Additive Inverse: $a + (-a) = 0$

3. Multiplication is associative: $(a \times b) \times c = a \times (b \times c)$

4. Distributive law: $a(b + c) = ab + ac$

Field: It is a special type of ring where division is possible

Properties of field

1. All ring properties hold.

2. Multiplication is commutative: $a \cdot b = b \cdot a$
3. Multiplicative Identity exists: $a \cdot 1 = a$
4. Multiplicative Inverse exists: $a \cdot a^{-1} = 1$

Prime & Relative Prime

Prime Number: It is a natural number greater than 1 which has only two factors those are 1 and itself.

Ex: 2, 3, 5, 7, 11, 13, 17, ...

→ 2 is the smallest and only even prime number

→ Numbers having more than two factors are called composite numbers.

→ Prime numbers are used in cryptography and key generation.

Relative Prime: Two numbers are called relative prime if their greatest common divisor (GCD) is 1.

Ex: 2, 3

→ These are important in Modular Arithmetic and RSA algorithm

Modular arithmetic: It is a system of arithmetic where numbers wrap around after reaching a certain value called modulus.

$$a \equiv b \pmod{n}$$

Ex:

$$17 \bmod 5 = 2$$

→ Used in cryptographic algorithms like RSA

→ Helps perform calculations with very large numbers efficiently.

Primality testing:

It is the process of checking whether a given integer n is prime or composite.

→ It plays a key role in cryptography.

→ Algorithm such as RSA heavily based on primality test.

Types of Primality testing:

1. Naive Algorithm

2. Miller Rabin Primality test

Naive Algorithm

It is also called Trial Division to check whether a number n is prime or not by testing divisibility with integers from 2 to $\sqrt{n}$.

Algorithm

1. Input a number n

2. If $n \leq 1$, it is not prime

3. for $i = 2$ to $\sqrt{n}$:

if $n \bmod i = 0$, then n is composite.

4. If no divisor is found, n is prime

Ex: $n = 19$

$$\sqrt{19} \approx 4.3$$

we need to check whether 19 is divisible 2, 3, 4

It is not divisible by them so it is
prime number.

Drawback: very slow for large numbers.

Miller-Rabin Primality test

→ It is a probabilistic algorithm used to check whether a number is prime.

Algorithm

1. $n-1 = 2^k \times m$

2. choose a random number a such that $1 < a < n-1$

3. compute $b = a^m \bmod n, \dots, b_i = b_{i-1}^2 \bmod n$

If $+1 \rightarrow$ composite

$-1 \rightarrow$ Probably Prime

Ex: Checking 561 is Prime or not

1. $n = 561$

$$n-1 = 560 = 2^4 \times 35 \Rightarrow k=4, m=35$$

2. choosing $a=2$ then

$$1 < 2 < 560 \checkmark$$

$$b_0 = a^m \bmod n = 2^{35} \bmod 561 = 263$$

$$b_1 = b_0^2 \bmod n = (263)^2 \bmod 561 = 166$$

$$b_2 = b_1^2 \bmod n = (166)^2 \bmod 561 = 67$$

$$b_3 = b_2^2 \bmod n = (67)^2 \bmod 561 = 1$$

here, $b_3 = 1$

So, it is a composite number.

Fermat's & Euler's Theorem

Fermat's Little Theorem: If p is a prime number and a is positive integer not divisible by p then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Ex: Let $p=5, a=2$

Given, $P=5, a=2$

$$a^{p-1} \equiv 1 \pmod{p}$$

$\therefore$ It is true for $P=5$
and $a=2$

$$2^4 \equiv 1 \pmod{5}$$

$$16 \equiv 1 \pmod{5} \checkmark$$

2) Let $P=13, a=11$

$$a^{p-1} \equiv 1 \pmod{p}$$

$$a^{12} \equiv 1 \pmod{13}$$

$$\Rightarrow (3)^3 \equiv 1 \pmod{13}$$

$$11^{12} \equiv 1 \pmod{13}$$

$$27 \equiv 1 \pmod{13} \checkmark$$

$$-2^{12} \equiv 1 \pmod{13}$$

$$-2^{4 \times 3} \equiv 1 \pmod{13}$$

$\therefore$ It is true for $P=13$ and
 $a=11$.

$$(-2^4)^3 \equiv 1 \pmod{13}$$

$$(16)^3 \equiv 1 \pmod{13}$$

Euler's Theorem

For every positive integer, 'a' & 'n' which are
said to be relatively prime, then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

where $\phi(n)$ is called Euler's totient function

Properties

1. $\phi(1) = 0$
2. $\phi(p) = p-1$ if p is a prime
3. $\phi(m \times n) = \phi(m) \times \phi(n)$ if m, n are relatively prime
4. $\phi(p^e) = p^e - p^{e-1}$ if p is a prime

5. If $n = p_1^{e_1} \times p_2^{e_2} \times \dots \times p_k^{e_k}$

$$\phi(n) = (p_1^{e_1} - p_1^{e_1-1}) \times (p_2^{e_2} - p_2^{e_2-1}) \times \dots \times (p_k^{e_k} - p_k^{e_k-1})$$

Examples:- $\phi(13) = \phi(13) = 13-1 = 12$ ($\because 13$ is prime)

$$\phi(10) = \phi(2) \times \phi(5) = (2-1)(5-1) = 4 \quad \left(\because 2, 5 \text{ are relatively prime} \right)$$

$$\phi(240) = 2^4 \times 3^1 \times 5^1 = (2^4 - 2^3)(3^1 - 3^0)(5^1 - 5^0) = 64$$

Prove Euler's theorem hold true for $a=3$ and $n=10$

$$a=3, n=10$$

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

$$\phi(10) = 4$$

$$3^{\phi(10)} \equiv 1 \pmod{10} \Rightarrow 3^4 \equiv 1 \pmod{10}$$

$$81 \equiv 1 \pmod{10} \checkmark$$

Euclidean Algorithm: It is an efficient method to compute greatest common divisor of two integers.

$$\text{gcd}(12, 33)$$

[$\because$ If B is 0 the A is the gcd]

Q	A	B	R
2	33	12	9
1	12	9	3
3	9	3	0
X	3	0	X

i) Biggest number should be placed in A

$$\therefore \text{gcd}(12, 33) = 3$$

$$\text{GCD}(750, 900)$$

Q	A	B	R
1	900	750	150
5	750	150	0
X	150	0	X

$$\therefore \text{GCD}(750, 900) \text{ is } 150$$

Algorithm

Prerequisite: $a > b$

Euclid - $\text{GCD}(a, b)$:

if $b = 0$ then
return a ;

else
return Euclid - $\text{GCD}(b, a \bmod b)$;

$$\text{Ex: } \text{GCD}(50, 12)$$

$$\text{GCD}(a, b) = \text{GCD}(b, a \bmod b)$$

$$= \text{GCD}(12, 50 \bmod 12)$$

$$= \text{GCD}(12, 2)$$

$$= \text{GCD}(2, 12 \bmod 2)$$

$$= \text{GCD}(2, 0)$$

$$= 2$$

Primitive root: ' α ' is said to be primitive root of prime number ' p ', if $\alpha \bmod p, \alpha^2 \bmod p, \dots, \alpha^{p-1} \bmod p$ are distinct

Ex: Is 2 primitive root of 5?

$$2^1 \bmod 5 = 2$$

$$2^2 \bmod 5 = 4$$

$$2^3 \bmod 5 = 3$$

$$2^4 \bmod 5 = 1$$

$$\Rightarrow 1, 2, 3, 4$$

distinct values

So 2 is primitive root

Extended - Euclidean Algorithm

1) $A > B$

2) $T_1 = 0, T_2 = 1$

3) $T = T_1 - (T_2 \times Q)$

4) T_1 is the multiplicative Inverse

What is the multiplicative inverse of 3 ^{mod} 5?

Q	A	B	R	T_1	T_2	T
1	5	3	2	0	1	-1
1	3	2	1	1	-1	2
2	2	1	0	-1	2	-5
X	1	0	X	2	-5	X

$1 - (-1)(2)$

It is the Multiplicative inverse

$\therefore 2$ is the M-I of 3 mod 5

Chinese Remainder Theorem

It is used to solve a set of different congruent equations with one variable but different moduli which are relatively prime

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$\vdots$

$$x \equiv a_n \pmod{m_n}$$

It states that above equations have a unique solution if the moduli are relatively prime

$$x = (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + \dots + a_n M_n M_n^{-1}) \bmod M$$

where,

$$M = m_1 \times m_2 \times m_3 \times \dots \times m_n$$

$$M_1 = \frac{M}{m_1}, M_2 = \frac{M}{m_2}, \dots$$

To find m_i^{-1}

$$M_1 M_1^{-1} \equiv 1 \bmod m_1$$

$$M_n M_n^{-1} \equiv 1 \bmod m_n$$

Example Solve using CRT

$$x \equiv 2 \bmod 3$$

$$x \equiv 3 \bmod 5$$

$$x \equiv 2 \bmod 7$$

Sol:

$$a_1 = 2, m_1 = 3$$

$$a_2 = 3, m_2 = 5$$

$$a_3 = 2, m_3 = 7$$

$$M = m_1 \times m_2 \times m_3$$

$$= 3 \times 5 \times 7$$

$$\boxed{M = 105}$$

$$M_1 = \frac{M}{m_1} = \frac{105}{3} = 35; M_2 = \frac{M}{m_2} = \frac{105}{5} = 21$$

$$M_3 = \frac{M}{m_3} = \frac{105}{7} = 15$$

$$M_1 \times M_1^{-1} \equiv 1 \bmod m_1$$

$$35 \times M_1^{-1} \equiv 1 \bmod 3 \Rightarrow M_1^{-1} = 2$$

lly

$$M_2^{-1} = 1$$

$$M_3^{-1} = 1$$

$$x \equiv (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + a_3 M_3 M_3^{-1}) \pmod{M}$$

$$= (2(35)(2) + 3(21)(1) + 2(15)(1)) \pmod{105}$$

$$= 233 \pmod{105}$$

$$\boxed{x = 23}$$

To check x is correct or not, we check in the following way

$$23 \equiv 2 \pmod{3} \checkmark$$

$$23 \equiv 3 \pmod{5} \checkmark$$

$$23 \equiv 2 \pmod{7} \checkmark$$

Example 2: Solve the following using CRT

$$4x \equiv 5 \pmod{9}$$

$$2x \equiv 6 \pmod{20}$$

Sol: $4x \equiv 5 \pmod{9}$
multiply by 4^{-1} on both sides

$$4 \times 4^{-1} x \equiv 4^{-1} \times 5 \pmod{9}$$

$$x \equiv (4^{-1} \pmod{9}) \times 5 \pmod{9}$$

$$\equiv 7 \times 5 \pmod{9}$$

$$\equiv 35 \pmod{9}$$

$$\boxed{x \equiv 8 \pmod{9}} \rightarrow \textcircled{1}$$

$$2x \equiv 6 \pmod{20}$$

$$\cancel{2}x \equiv \cancel{2} \times 3 \pmod{20}$$

$$\boxed{x \equiv 3 \pmod{20}} \rightarrow \textcircled{2}$$

$$4^{-1} \pmod{9}$$

$\downarrow$

it means that when we do $a \pmod{b}$ the value should be 1

here if we take 7

$$4(7) \pmod{9}$$

$$= 1$$

$$\therefore 4^{-1} \pmod{9} = 7$$

$\Rightarrow$ Now for eq's

$\textcircled{1}$ & $\textcircled{2}$ we

apply CRT

then we get

$$x = 143$$

Discrete Logarithm

It is in the form $g^x \equiv y \pmod{P}$

where, P is prime

$x \rightarrow$ unknown exponent

Solve $\log_2 9 \pmod{11}$

$P=11$, $g=2$ (base), $X=9$

$$\log_g x \equiv n \pmod{P}$$

$$x \equiv g^n \pmod{P}$$

$$9 \equiv 2^n \pmod{11}$$

for which value of n 9 is the congruent to $2^n \pmod{11}$

for $n=6$

$9 \equiv 64 \pmod{11} \Rightarrow$ satisfies

$$\boxed{\therefore n=6}$$